GNBF5020

Lecture Quick Reference

講次速查表

The five lectures in one document: what you must be able to do in each lecture, and every exact number Dr. Wong stated.

GNBF5020 · Introduction to Molecular Biology and Genetics
Dr. Tony Wong · Lectures 1-5, 12 Sep - 17 Oct 2026

> **[ENGLISH ONLY]** The exam must be answered in English. Dr. Wong said this three separate times, and once he said he gave zero marks for it. This document is in English so you can revise directly in the language you will be graded in. The Chinese companion exists only for explanation; gene, protein, pathway, method and drug names stay in English in both versions.

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The five lectures in one document: what you must be able to do, and the numbers he gave

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> ***How this differs from the other documents.*** *The Beginner's Guide teaches you the content from zero. The Midterm Drill Pack trains you on the exam format. This document is the* ***one-page-per-lecture reference****: for each lecture, the list of things you must be able to do, plus every exact number he stated, because he said every number is exam content.*

Lecture 1 — Introduction to Biomolecules

You must be able to: - State the central dogma and name its exceptions (reverse transcriptase, RNA replication, prions) - Distinguish the four biomolecule classes and give one function of each - Draw a nucleotide and a peptide: label 1′/3′/5′, phosphate, sugar, base - Explain why DNA uses thymine rather than uracil - Describe primary → secondary → tertiary → quaternary structure - Explain allostery, and why a chaperone matters - Explain why hydrogen bonds and base stacking both matter for DNA stability

The numbers and facts he gave

Item	What it is	Why it matters
4	The four biomolecule classes: carbohydrate, lipid, protein, nucleic acid	The skeleton of the whole course
3	Nucleotide components: phosphate, sugar, base	Draw it correctly
2	Hydrogen bonds in A–T; 3 in G≡C	DNA structure questions
20	Standard amino acids	Every MCQ with an amino acid
1.65 turns / ~146 bp	DNA wrapped per nucleosome	Chromatin structure
≥ 200 nt	RNA length above which it is a long non-coding RNA (lncRNA)	Lect 5
22 nt	Length of a microRNA	Lect 5

Common traps - Lipids are **not** polymers. - Peptide bonds join amino acids; phosphodiester bonds join nucleotides; glycosidic bonds join sugars. - Unambiguous ≠ specific. The code is degenerate **and** unambiguous.

Lecture 2 — Basic Cell Biology

You must be able to: - Compare prokaryote and eukaryote, with the endosymbiotic evidence - Name the organelles and say what each does - Draw the fluid mosaic model and label phospholipid bilayer, protein, carbohydrate, cholesterol - Distinguish simple diffusion, facilitated diffusion, osmosis, and active transport - Predict what happens to a

cell in a hypertonic, hypotonic, or isotonic solution - Lay out G1 → S → G2 → M → cytokinesis and say what happens in each - Distinguish mitosis from meiosis I, and say what crossing over and independent assortment do - Read a pedigree and identify the inheritance pattern

The numbers and facts he gave

Item	What it is	Why it matters
0.1–0.4 µm	E. coli size	Prokaryote scale
20–30 µm	Typical mammalian cell	Eukaryote scale
2 membranes	Mitochondria and chloroplasts	Endosymbiotic evidence
70S / 80S	Prokaryote / eukaryote ribosome	Why some antibiotics are safe
23 pairs (46)	Human chromosomes	Genome structure
V ≈ 39, D = 27, J = 6	V(D)J gene segment copy numbers	Lect 4, and SCID

Mitochondrial inheritance — the three hallmarks 1. An affected mother passes the condition to **all** her children 2. An affected father passes it to **none** 3. Only females transmit it onward

The trap he flagged: mitochondrial *disease* can be caused by a mutation in **nuclear** DNA too, and then it inherits in a Mendelian way. Do not read "mitochondrial" off the pedigree alone — ask which gene.

Lecture 3 — DNA and Chromatin Structure

You must be able to: - State the experiments proving DNA is the genetic material (Griffith, Avery, Hershey–Chase) - Describe Watson–Crick: antiparallel, complementary, base pairs, grooves - Draw a nucleotide, a phosphodiester linkage, and the double helix - Define nucleosome, chromatin, euchromatin, heterochromatin, and the packing levels - Explain what a histone mark does, and distinguish activating from repressive marks - Define centromere, telomere, and origin of replication - Explain methylation and how it is measured (bisulfite sequencing)

The numbers and facts he gave

Item	What it is	Why it matters
1953	Watson and Crick	The model
1958	Meselson and Stahl	Proved semi-conservative replication
1962	Nobel Prize (Watson, Crick, Wilkins)	Context
146 bp	DNA per nucleosome core	Chromatin
1.65 turns	Turns of DNA per nucleosome	Chromatin
5–8 nm → 700 nm	Chromatin fibre diameter range	Packing levels
30 nm	The (classical) solenoid fibre	Packing levels
200 nt	RNA length above which it is an lncRNA	Lect 5
425 and 250 bp	The two bands in the liver-cancer splicing gel	Lect 5

Common traps - Antiparallel, not parallel. This one answer explains leading vs. lagging. - Major groove is the **wider** one, and it is where sequence-specific proteins read DNA. - Promoter methylation **represses**; demethylation activates. - Bisulfite converts **unmethylated** C to uracil and leaves methylated C alone. The reverse is the classic wrong answer.

Lecture 4 — DNA Replication, Repair and Recombination

You must be able to: - State the Meselson–Stahl result after one and two generations - Name every enzyme at the replication fork and say what it does - Explain why the leading strand is continuous and the lagging strand is not - Describe the end-replication problem and how telomerase solves it - Name all six repair pathways and match each to the lesion it handles - Explain why NHEJ failure causes SCID - Explain homologous recombination, site-specific recombination, and transposition - Identify mutagens by name and say what lesion each produces

The numbers and facts he gave

Item	What it is	Why it matters
30%	Midterm's share of the final grade	Lect 1, but plan the midterm around it
V ≈ 39, D = 27, J = 6	V(D)J segment copy numbers	SCID chain
39 / 27 / 6	The raw material for immune repertoire diversity	SCID chain
15%	Share of colorectal cancers that are MSI-high	MMR failure → hot tumour
3 or more	HBV integration sites → significantly poorer survival	Lect 2
Top 10%	Longest telomeres in hepatocellular carcinoma had **worse** recurrence-free survival	Counter-intuitive; worth remembering
39.6	Odds ratio, CIMP-high vs CIMP-0 (95% CI 20.6–76.1, $p < 0.0001$)	Lect 3
12.7	Odds ratio for detected methylation (CI 7.84–20.7, $p < 0.0001$)	Lect 3

The six repair pathways

Pathway	Lesion it handles	Disease when defective
Direct reversal	One specific chemical lesion (photolyase for UV dimers; MGMT for O6-methylguanine)	— (humans lack photolyase)
BER	A single damaged base	MUTYH-associated polyposis
NER	Bulky lesions spanning several nucleotides	**Xeroderma pigmentosum**
MMR	Replication errors, small insertion-deletion loops	**Lynch syndrome (HNPCC)**
HR	Double-strand breaks, with a homologous template	**BRCA1/BRCA2** → breast, ovarian, prostate

Pathway	Lesion it handles	Disease when defective
NHEJ	Double-strand breaks, no template	Some SCID

Common traps - **BER vs NER:** one base versus a whole patch. NER failure gives you sunburn — that is your discriminator. - **HR vs NHEJ:** accurate with a template versus fast and error-prone without one. - UV makes **pyrimidine dimers**. EMS **alkylates** (G→A transitions). Ethidium bromide **intercalates** (frameshifts). Aflatoxin B1 gives **G→T transversions**. - Deamination of **5-methylcytosine** → thymine is silent and unrepairable, because it looks like a normal base.

Lecture 5 — Control of Gene Expression

You must be able to: - Explain the lac operon with **both** conditions and name the molecule responsible for each - Explain the trp operon as the mirror image, and define corepressor - Define cis and trans, and give the chromosome 1 / chromosome 13 example - List the levels at which eukaryotic gene expression is controlled - Explain alternative splicing and why 95% of human multi-exon genes use it - Explain epigenetic memory and how a differentiated cell stays differentiated - Explain microRNA, lncRNA, and the difference between them - Explain codon optimisation and why CHO cells need it - State the NF-κB cytokine time course

The numbers and facts he gave

Item	What it is	Why it matters
2 conditions	Lactose present **and** glucose low	The classic wrong answer is naming only lactose
3 genes	lacZ, lacY, lacA	One polycistronic message
5 genes	trpE, trpD, trpC, trpB, trpA	Switched off when Trp is plentiful
100 μM	Optimal IPTG concentration (range tested 0–1000 μM)	Recombinant production
200 nt	Length above which an RNA is an lncRNA	vs. small RNA
22 nt	Length of a microRNA	Binds a seed sequence
425 and 250 bp	Two bands in the liver-cancer splicing gel	Alternative splicing
23 and 19	Two patient groups: without vs. with alternative splicing	$P < 0.001$ on the log-rank test
220 and 97	microRNAs screened, and paired tumour / adjacent non-tumour tissues	Liver cancer screen
14q32.2	The over-expressed microRNA cluster	$P = 4.23 \times 10^{-10}$ vs log10 AFP; survival $P = 0.009$, $\chi^2 = 6.81$
Chromosome 1 / 13	A repressor on chr 1 acting on a gene on chr 13	The clearest trans-regulation example
4–6 hours	When TNF-α peaks	First cytokine of the NF-κB response

Item	What it is	Why it matters
Order	TNF-α → IL-1β → IFN-γ → IL-2 and IL-15	Know the shape and the time course

The lac operon, precisely

Condition	Mechanism	Molecule
Lactose present	Allolactose binds the Lac repressor and removes it from the operator	**Allolactose** (made by lacZ-encoded β-galactosidase — **not** lactose itself)
Glucose low	cAMP rises; the cAMP–CAP complex binds the promoter and recruits the copying machinery	**cAMP–CAP**

The trp operon: tryptophan binds the Trp repressor as a **corepressor**; the complex binds the operator and switches tryptophan biosynthesis off. **IPTG** is the artificial, non-metabolised inducer used in *E. coli* recombinant systems.

Common traps - Naming only “lactose” for the lac operon — half marks at best. - Calling a repressor *cis* and a factor *trans* backwards: *cis* = DNA on the same molecule; *trans* = a diffusible factor. - NF-κB: the cytokine **list** is not the exam point; the **sequence and the time course** are.

Quick self-test

Tick a box only if you can do it **today, unaided**.

- ☐ I can write the central dogma and name three exceptions.
- ☐ I can label a nucleotide and a peptide.
- ☐ I can say why DNA uses thymine.
- ☐ I can distinguish the four biomolecule classes and their functions.
- ☐ I can name the organelles and their jobs.
- ☐ I can predict a cell’s fate in a hypertonic solution and say why.
- ☐ I can lay out the cell cycle and the stages of mitosis.
- ☐ I can name the three hallmarks of mitochondrial inheritance and the nuclear-DNA trap.
- ☐ I can draw a nucleosome and describe the packing levels.
- ☐ I can name the Meselson–Stahl result and every enzyme at the fork.
- ☐ I can explain why the lagging strand must be discontinuous.
- ☐ I can match all six repair pathways to their lesions.
- ☐ I can draw the NHEJ → V(D)J → SCID chain with the copy numbers.
- ☐ I can state both lac operon conditions and the molecule for each.
- ☐ I can name the three lac genes and the five trp genes.
- ☐ I can define cis and trans with the chromosome 1 / 13 example.
- ☐ I can explain alternative splicing with the 425 / 250 bp bands.
- ☐ I can state the microRNA length, the lncRNA cutoff, and the IPTG optimum.
- ☐ I can say why CHO cells need codon optimisation.

- ☐ I can state the NF-κB cytokine order and the TNF-α peak time.

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